第25小时

专业英语

25.0 章节考点分析

第 25 小时主要学习专业英语知识。根据考试大纲，上午单选题会有 5 道英文选择题（分值为 5 分），主要涉及信息技术与管理类的一些概念性的知识。这部分知识对于一些考生来说是难点所在，在本小时内容里总结出一些软考中常考的英文知识点供广大考生参考。

【导读小贴士】

系统架构设计师考试英文试题的出题范围基本局限于系统架构设计方面基础性的、概念性的知识，大多属于名词解释范畴。如果考生具有一定的英文水平，同时对于基本概念掌握得比较牢固，这部分分值不难拿到。

25.1 架构风格

An architectural style defines as a family of such systems in terms of a pattern of structural organization. More specifically，an architectural style defines a vocabulary of components and connector types，and a set of constraints on how they can be combined. For many styles there may also exist one or more semantic models that specify how to determine a system’s overall properties from the properties of its parts. Many of architectural styles have been developed over the years.

The best-known examples of pipe-and-filter architectures are programs written in the UNIX shell.

一种架构风格以一种结构化组织模式定义一组这样的系统。具体来说，一种架构风格定义了一个构件及连接器类型的词汇表，以及一组关于它们如何能够被关联的约束。对于许多风格来说，可能也存在一个或多个语义模型，从系统部件的特性来确定系统的整体特性。许多架构风格已经发展了很多年，众所周知的管道-过滤器架构就是用 UNIX shell 编写的程序。

25.2 非功能需求

【基础知识点】

The architecture design specifies the overall architecture and the placement of software and hardware that will be used. Architecture design is a very complex process that is often left to experienced architecture designers and consultants. The first step is to refine the nonfunctional requirements into more detailed requirements that are then employed to help select the architecture to be used and the software components to be placed on each device. In a client-based architecture, one also has to decide whether to use a two-tier, three-tier, or n-tier architecture. Then the requirements and the architecture design are used to develop the hardware and software specification. There are four primary types of nonfunctional requirements that can be important in designing the architecture. A operational requirements specify the operating environment(s) in which the system must perform and how those may change over time. Performance requirements focus on the nonfunctional requirements issues such as response time, capacity, and reliability. Security requirements are the abilities to protect the information system from disruption and data loss, whether caused by an intentional act. Cultural and political requirements are specific to the countries in which the system will be used.

架构设计指定了将要使用的软件和硬件的总体架构和布局。架构设计是一个非常复杂的过程，往往留给经验丰富的架构设计师和顾问。第一步是将非功能需求细化为更详细的要求，然后用于帮助选择要使用的体系结构以及要放置在每个设备上的软件组件。在基于客户端的架构中，还必须决定是使用两层、三层还是 *n* 层架构。然后使用需求和体系结构设计来开发硬件和软件规范。有 4 种主要的非功能需求类型可能在设计架构时非常重要。操作要求指定系统必须执行的操作环境以及这些操作环境如何随时间变化。性能要求侧重于非功能性需求问题，如响应时间、容量和可靠性。安全要求是指是否有能力保护信息系统免受故意行为造成的破坏和数据丢失。文化和政治要求明确了特定系统将被使用的国家。

25.3 应用架构

【基础知识点】

An application architecture specifies the technologies to be used to implement one or more information systems. It serves as an outline for detailed design, construction, and implementation. Given

the models and details, include logical DFD and ERD, we can distribute data and processes to create a general design of application architecture. The design will normally be constrained by architecture standards，project objectives, and the feasibility of techniques used. The first physical DFD to be drawn is the network architecture DFD. The next step is to distribute data stores to different processors. Data partitioning and replication are two types of distributed data which most RDBMSs support. There are many distribution options used in data distribution. In the case of storing specific tables on different servers we should record each table as a data store on the physical DFD and connect each to the appropriate server.

应用架构说明了实现一个或多个信息系统所使用的技术，它作为详细设计、构造和实现的一个大纲。通过给定的包括逻辑数据流图和实体联系图在内的模型和详细资料，我们可以分配数据和过程以创建应用架构的一个概要设计。概要设计通常会受到架构标准、项目目标和所使用技术的可行性的制约。需要绘制的第一个物理数据流图是网络架构数据流图。接下来是分配数据存储到不同的处理器。数据分区和复制是大多数关系型数据库支持的两种分布式数据形式。有许多分配方法用于数据分布。在不同服务器上存储特定表的情况下，我们应该将每个表记为物理数据流图中的一个数据存储，并将其连接到相应的服务器。

25.4 软件架构重用

【基础知识点】

Software architecture reconstruction is an interpretive, interactive, and iterative process including many activities. Information extraction involves analyzing a system's existing design and implementation artifacts to construct a model of it. The result is used in the following activities to construct a view of the system. The database construction activity converts the elements and relations contained in the view into a standard format for storage in a database. The view fusion activity involves defining and manipulating the information stored in database to reconcile, augment, and establish connections between the elements. Reconstruction consists of two primary activities: visualization and interaction, pattern definition and recognition. The former provides a mechanism for the user to manipulate architectural elements, and the latter provides facilities for architecture reconstruction.

软件架构重用是一个解释性、交互式和反复迭代的过程，包括了多项活动。信息提取通过分析系统现有设计和实现工件来构造它的模型。其结果用于在后续活动中构造系统的视图。数据库构建活动把视图中包含的元素和关系转换为数据库中的标准存储格式。视图融合活动包括定义和操作数据库中存储的信息，理顺、加强并建立起元素之间的连接。重构由两个主要活动组成：可视化和交互式及模式定义与识别。前者提供了一种让用户操作架构元素的机制，后者则提供了用于架构重构的设施。

25.5 练习题

A system's architecture is a representation of a system in which there is a mapping of__（1）__onto hardware and software components, a mapping of the__（2）__onto the hardware architecture, and a concern for the human interaction with these components. That is, system architecture is concerned with a total system, including hardware, software, and humans.

Software architectural structures can be divided into three major categories, depending on the broad nature of the elements they show.__（3）__embody decisions as a set of code or data units that have to be constructed or procured.__（4）__embody decisions as to how the system is to be structured as set of elements that have run-time behavior and interactions.__（5）__embody decisions as to how the system will relate to non-software structures in its environment (such as CPUs, file systems, networks, development teams, etc.).

（1）A．attributes　B．constraint　C．functionality　D．requirements

（2）A．physical components　B．network architecture
C．software architecture　D．interface architecture

（3）A．Service structures　B．Module structures
C．Deployment structures　D．Work assignment structures

（4）A．Decompostion structures　B．Layer structures
C．Implementation structures　D．Component and connector structures

（5）A．Allocation structures　B．Class structures
C．Concurrency structures　D．Uses structures

解析：系统架构是一个系统的一种表示，包含了功能到软硬件构件的映射、软件架构到硬件架构的映射以及对于这些组件人机交互的关注。也就是说，系统架构关注于整个系统，包括硬件、软件和使用者。

软件架构结构根据其所展示元素的广义性质，可以分为 3 个主要类别。

（1）模块结构将决策体现为一组需要被构建或采购的代码或数据单元。

（2）构件连接器结构将决策体现为系统如何被结构化为一组具有运行时行为和交互的元素。

（3）分配结构将决策体现为系统如何在其环境中关联到非软件结构，如 CPU、文件系统、网络、开发团队等。

答案：（1）C　（2）C　（3）B　（4）D　（5）A